

Toward Readable Long Exact Palindromes

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Abstract

Our central question is whether a novel exact palindrome can become longer without ceasing to read as English. Exactness and length are mechanically verifiable, but neither is a language-quality result: every chosen letter fixes another letter in reverse, while word boundaries may move. We therefore treat grammaticality, identifiable subject or intent, and whole-text coherence as the primary outcomes. We freeze a blinded 30-item human study of a novel exact composer at three lengths (roughly 60, 230, and 460 letters), with length-matched real-prose and shuffled-word controls and catalogue references. The packet is frozen before independent ratings; it does not yet establish that the system produces readable long prose. Two diagnostics explain why the evaluation is necessary. Across 150 WikiText-2 spans at each of six lengths, three segmenters, and two causal language models, reverse re-segmentation adds 2.14–3.34 bits per letter and reduces multi-letter dictionary coverage from 86.9–91.5% to 49.6–54.6%. A POS-shape gate illustrates that a formal structural filter changes which exact candidates a bounded search reaches, but does not measure English quality. Current long artifacts meet exactness and length constraints, but have not been validated as prose. The paper’s claim is therefore evaluative: readable long palindromes require blinded reader evidence, not a larger search count or automatic score.

1 Introduction

Let $n(x)$ lowercase a text and retain only ASCII letters. A text x is palindromic when

$$n(x) = \text{rev}(n(x)). \quad (1)$$

Constructing one from words requires coordinating choices on both sides: once the left half’s letters are chosen, the right half is fixed in reverse. The two halves may still place spaces differently.

The phrase pair *step on* $\leftrightarrow$ *no pets*, for example, forms the exact palindrome “Step on no pets.” We call such halves a *mirror pair*: two word sequences whose normalized letter strings reverse to one another. Concatenating a mirror pair produces an exact palindrome even when its two halves use different word boundaries.

The scientific target is the longest *novel* exact palindrome that still reads as coherent English. “Longer” is valuable only after a text clears this reading requirement; a dictionary list can be extremely long and still be a worse language-generation result than a short, coherent palindrome. We use three reader outcomes: grammaticality, whether a reader can identify a subject or intent, and whole-text coherence. Exactness is a fourth, deterministic property rather than a proxy for any of them.

The present paper first measures a narrower property. Starting from ordinary English, we remove formatting, re-segment the resulting letters in both directions with the same vocabulary and objective, and score both versions with the same causal language model. This avoids charging reversal for the separate cost of replacing the author’s casing, punctuation, and word boundaries. With the corpus samples, vocabulary, segmenter, and scorer fixed, how much more does the reversed lexicalization cost?

Our primary empirical object is a frozen blinded reader study: nine novel system outputs at three requested lengths, matched ordinary-English and shuffled-word controls, and catalogue references. Its source key is withheld from readers; three independent raters will score grammaticality, identifiable subject or intent, and coherence on separate 0–3 scales. The study is frozen, but no ratings are available at the time of writing, so this manuscript makes no positive readability claim. The paired reversal measurement and the exact decoder with its structural filter are supporting diagnostics: they

describe the language burden imposed by reversal and the source of candidate material. Sections 2–3 report the former; Section 4 reports the latter; Section 5 defines the primary outcome.

2 Paired reversal measurement

Let $z = n(x)$ be the lowercase ASCII letter sequence of an English span and let s segment a letter stream using a fixed vocabulary V . Single letters are legal fallbacks, so every stream is segmentable. We apply the same s to z and $\text{rev}(z)$. Let $\tau_M(y) = (t_1, \dots, t_k)$ be model M ’s tokens for a segmented string y , and let b_M be one shared model-defined boundary token. We score every text token and normalize by letters:

$$H_M(y; z) = -\frac{1}{|z|} \sum_{j=1}^k \log_2 p_M(t_j \mid b_M, t_{<j}). \quad (2)$$

The *reversal cost* is the paired difference

$$C_{M,s}(z) = H_M(\text{seg}_s(\text{rev}(z)); z) - H_M(\text{seg}_s(z); z). \quad (3)$$

A positive value means that the reversed letters are costlier after both directions receive the same segmentation treatment. This is autoregressive LM surprisal, not an information-theoretic bound or a human judgment.

2.1 Frozen design

We sample spans at word boundaries from the first four million characters of the WikiText-2 raw training split (Merity et al., 2016). Section headings and sentence fragments shorter than 40 characters are excluded. For each target of 20, 30, 40, 60, 80, or 120 letters, a seeded sampler saves 150 spans of at least that length; no sample is filtered using its reversal. The same saved spans are reused in every model–segmenter condition at that target. For each span and segmenter, we score the naturally formatted source, the automatically segmented forward stream, and the automatically segmented reverse stream. The primary comparison uses the latter two; the natural source is a control that isolates the cost of replacing the author’s formatting.

The source lexicon has 52,927 lines. ASCII and short-word checks retain 52,793 entries of at most 20 letters. Coverage is the fraction of letters placed in lexicon words of at least three letters, excluding the fallbacks that make segmentation total. We test

Segmenter	Forward cov.	Reverse cov.	Cost range
Unigram	88.7–90.0	49.6–51.3	2.90–3.34
Fewest	90.4–91.5	52.3–54.6	2.79–3.23
Greedy	86.9–88.3	49.9–52.8	2.14–2.71

Table 1: Ranges over two models and six target lengths. Coverage is percent of letters inside lexicon words of length at least three; cost is additional bits per letter after reversal.

three deliberately different objectives: UNIGRAM maximizes summed Zipf frequency (Speer, 2022) minus a fixed word penalty; FEWEST minimizes segment count; and GREEDY selects the longest available word left-to-right. These are sensitivity conditions, not models of human word recognition.

GPT-2 (124M) (Radford et al., 2019) and SmolLM2-135M (Ben Allal et al., 2025) each score all 18 length–segmenter cells, giving 36 conditions in total. They are similar in size but differ in architecture, tokenizer, and training pipeline. Exact model, corpus, vocabulary, and software revisions are stored with the results.

3 Results

Reversal increases cost in every condition (Figure 1). The 36 means range from 2.14 to 3.34 additional bits per letter, with standard errors from 0.04 to 0.11. Segmentation changes the magnitude but not the direction (Table 1). The largest gap between the two models in a matched length–segmenter condition is 0.136 bits per letter, so the result is not specific to one of these model families.

The model-free coverage diagnostic agrees with the likelihood result: the forward stream places roughly nine in ten letters in multi-letter words, whereas the reversal places roughly half. Coverage alone is not readability; a fully covered sequence may still be nonsensical.

The natural-formatting control also matters. Replacing original formatting with an automatic forward segmentation adds 0.79–1.73 bits per letter across conditions. A natural-text-versus-reversal comparison would fold this penalty into reversal. Pairing automatically segmented forward and reverse streams removes that confound, at the cost of measuring the chosen segmentation procedures rather than the author’s boundaries.

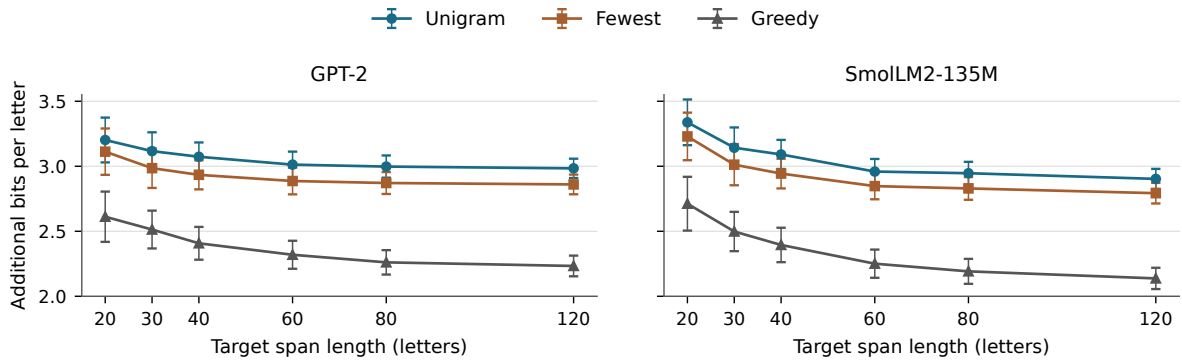


Figure 1: Mean paired reversal cost across 150 spans per condition; error bars are normal 95% intervals computed from the within-condition paired standard error. Every text token is scored after a shared boundary token.

3.1 Magnitude, length, and model agreement

The paired difference is computed per span before averaging, so a long span does not receive more weight than a short span within a condition. In perplexity-ratio terms, the observed cost corresponds to $2^C = 4.4\text{--}10.1$ per letter. This restatement does not mean that independent letters survive with that probability: letters, tokens, and segmentations are strongly dependent, and the experiment never samples palindromes. It only describes how much more model surprisal the same letter material acquires in the reverse direction.

All six model-segmenter series decline between the 20- and 120-letter targets. At 20 letters the means span 2.61–3.34 bits per letter; at 120 they span 2.14–2.98. One plausible contributor is dilution of the shared initial boundary as spans grow. The current design does not identify a length law: target-length samples share a seeded sentence order, and sentence and token effects are correlated. We therefore report the curve descriptively rather than fitting or extrapolating it.

The two models agree more closely with each other than the three segmenters do. Their maximum matched-cell difference is 0.136 bits per letter, whereas the range between unigram and greedy conditions exceeds half a bit at most lengths. The boundary-recovery rule is therefore part of what the experiment measures, not an innocuous preprocessing choice. This also motivates reporting coverage alongside the LM score.

3.2 A saved example

The first frozen 40-letter sample is “Scuba diving is increasingly popular in Ireland” (41 normalized letters). Under unigram segmentation, 85.4% of

its forward letters but 46.3% of its reversed letters lie inside lexicon words of at least three letters. The remaining reversal is tiled largely by short words and one-letter fallbacks. This trace illustrates why total segmentation is needed: dropping streams without a complete multi-letter analysis would select an unusually easy subset of reversed English. The release stores all samples, not only this example, and the audit re-runs each segmentation.

4 Exact candidate construction (not a quality result)

The measurement shows higher cost for reversed streams, but it does not say how many formally valid candidates a decoder can return or whether readers will understand them. We use an exact two-ended search derived from the remainder construction of Norvig (2016). The state stores left and right word sequences and the unmatched remainder. This remainder, or *overhang*, records the letters one side still owes the other. A trie returns words compatible with the overhang. Every closure is checked by the normalized letter predicate, so model ranking remains separate from validity.

For each lowercased word in the Brown Corpus, we retain its observed universal tags (Francis and Kučera, 1979; Bird et al., 2009; Petrov et al., 2012). From the corpus sentences, we retain 5,649 three-to-nine-tag patterns after removing punctuation and X tags and requiring a verb and an opening PRON, DET, NOUN, ADJ, NUM, or ADV. A word may have several observed tags. A partial left sequence must match the suffix of at least one retained pattern; a partial right sequence must match a prefix. Because later moves only

prepend on the left and append on the right, a failed match cannot be repaired by a descendant. This pruning preserves completions that satisfy the finite tag-pattern constraint. The pattern inventory is not a grammar, and an admitted sequence need not be meaningful.

The *terminal-only* arm applies the POS test only to complete mirror pairs. The *incremental* arm also rejects a partial state as soon as it cannot complete an allowed pattern; both arms retain the same final test. We use the 21,073 Brown-tagged words in a frequency-ranked 30,000-word list because the POS test requires observed tags. The filter is a reproducible way to restrict the source of candidate mirror pairs, not a grammar or a quality score.

Any bounded traversal reaches a policy-dependent subset of the exact state space. In a robustness check using 20 fixed openings, two child orders, and a 50,000-pop cap, incremental filtering produced more accepted pairs under both depth-first orders, while the breadth-first arms produced no terminal-only closures and only one incremental closure before the cap. Thus accepted-pair counts, generated children, visited states, and CPU time are not reported as evidence about language or as a general search advantage. The frozen reader study uses fixed outputs without language-quality postselection; its reader scores, rather than this construction diagnostic, decide whether additional length is valuable.

4.1 Why length is insufficient on current outputs

Given mirror pairs (L_i, R_i) and an optional palindromic centre C , the sequence $L_1 \dots L_k C R_k \dots R_1$ is palindromic. This identity permits assembly without another search to satisfy the letter mirror. We freeze and audit two such compositions. The generated example nests 24 distinct mirror pairs with no centre. It contains 101 words and 342 letters, and an independent audit verifies both exactness and that all 24 pairs came from the generated-pair bank rather than a known complete palindrome. The result is long enough to satisfy a mechanical 100-word target, but has no blinded reader evidence of a shared discourse. It is evidence that length can be assembled from novel exact units, not that the system writes a paragraph.

The catalogue example nests nine pairs around a centre, giving 72 words and 237 letters. Its units include familiar hand-written pairings such

as “lived on decaf” $\leftrightarrow$ “faced no devil.” The full construction is exact, but its sentences are borrowed from a palindrome catalogue. It is a useful presentation baseline and not a generation result. The third output comes from an adapted Norvig dictionary search that maximizes letter count under phrase-uniqueness and word-repetition restrictions. Its 90,937 letters are independently verified, but it is not evaluated as coherent prose. We report the three outputs separately because exact scale, generated material, and readable language are different outcomes. They motivate the primary reader study; none is evidence that the system has reached the target.

5 Primary outcome: blinded readability by length

We freeze a 30-item blinded packet before collecting ratings. It contains three novel, hierarchical exact compositions at each of three requested lengths (80, 240, and 480 letters; the rendered texts are approximately 58–60, 230–234, and 458–468 letters). Every system item uses a fixed seed and is retained without a language-quality score selecting it. Each has a length-matched ordinary-English passage and a shuffled-word version of that passage. Three short catalogue palindromes provide a human-written reference. The reader-facing packet contains only opaque identifiers and text; its source key is separate.

At least three independent raters score each item from 0 to 3 for grammaticality, recoverable subject or intent, and whole-text coherence. The primary comparison is coherence across the three system-length bands; the controls test whether the rubric distinguishes ordinary prose from word salad. We will report every rating, missing value, and ordinal inter-rater agreement. This protocol does not permit an automatic score, exactness check, or the search yield to decide readability. Ratings are pending, so the study is a frozen evaluation design rather than a result in this draft.

6 Related work and conclusion

Papadopoulos et al. (2015) conjoin forward and backward constraints in an n-gram graph and also use syntactic patterns. Lexically constrained decoding tracks outstanding requirements (Hokamp and Liu, 2017; Post and Vilar, 2018), while lookahead can represent future feasibility (Lu et al., 2022). Exact palindrome search combines both

problems: distant characters are coupled, but word boundaries may move under reversal.

The paired measurement establishes higher model cost and lower lexical coverage for reversed streams under the frozen conditions. The decoder experiment constructs valid candidates under its recorded stopping limit. Neither answers the paper’s primary question. The frozen reader study is designed to answer whether longer novel exact palindromes retain grammaticality, subject, and coherence; until independent ratings arrive, readable long prose is an unverified target rather than a claimed result.

Limitations

The measurement is English-only and uses one Wikipedia-derived corpus, one lexicon, ASCII normalization, three automatic segmenters, and two relatively small causal LMs. The numerical range may not transfer to larger or differently trained models. Training-corpus overlap is unknown. The six target-length samples are produced by the same seeded sentence order and are not independent replicates for estimating a length trend. Normal intervals summarize paired span variation within a condition; we do not treat the 36 condition means as independent observations. Neither lexicon coverage nor LM surprisal estimates human grammaticality or coherence, and the segmenters do not optimize the LM score.

The decoder study covers 50 selected opening subtrees from one vocabulary and one finite POS language. Its confidence intervals describe those paired searches, not a population of corpora or systems. The 90,937-letter artifact changes policy and restrictions relative to Norvig’s published construction; the 498-letter difference is an auditable artifact fact, not a performance comparison. The primary reader study is frozen but unrated: it has only three system texts per length band and does not yet have independent human responses. It cannot support a readability finding until those responses and their control separation are reported.

Ethical considerations

The work uses public text, dictionaries, and pre-trained models; no human participants or personal data were collected. The release retains source licenses and attribution rather than applying a blanket waiver. Dictionary generation can surface names, slurs, or other undesirable entries, and ex-

act outputs should not be presented as coherent language without inspection. The reported LM experiment uses two small models to limit compute.

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A Reproduction record

The evidence archive contains the exact sampled spans and normalized letters, all 36 aggregate rows, the 52,927-line source vocabulary, the two model commit hashes, corpus snapshot and file hash, package versions, experiment code, and an audit that recomputes design completeness, normalization, segmentation, coverage, and all reported arithmetic. From the archive root, run:

```
python3 -m \
  experiments.audit_mirror_cost \
  data/mirror-cost/results.json \
  -output mirror-cost-audit.json
```

This audit does not load model weights. Full logit reproduction requires the specified packages and cached/downloaded model revisions:

```
python3
  experiments/mirror_cost.py \
  -models gpt2 \
  HuggingFaceTB/SmolLM2-135M \
  -strategies unigram fewest
  greedy \
  -spans 20 30 40 60 80 120 -n 150
  \
  -seed 20260911 -batch-size 16 \
  -out rerun/results.json
```

The structural audit commands and exact input terms are listed in the archive README. The released PDF, source archive, evidence archive, and checksum manifest are deposited as an unpublished review draft.

The frozen blinded reader package accompanies the archived materials. It contains the opaque texts, instructions, and blank rating sheet; the source key is separate. After independent ratings are collected, the bundled analysis script validates the scores and reports each dimension and ordinal agreement without changing the item set.